Enrollment No.: ______________________

Darshan UNIVERSITY

Darshan Institute of Engineering & Technology
B.Tech. | Sem-4 | Summer-2024

Course Code	:	2101CS403	**Date**	:	19-04-2024
Course Name	:	Operating System	**Duration**	:	150 Minutes
			Total Marks	:	70

Instructions:
1. Attempt all the questions.
2. Figures to the right indicates maximum marks.
3. Make suitable assumptions wherever necessary.

Q.1 **(A)** Define the role of Operating System as a resource manager. **4**

(B) Explain Monolithic Kernel with the help of diagram. **3**

OR

Explain layered system with the help of diagram.

(C) Define operating system. Explain any five services/functions of operating system. **7**

OR

Define operating system. Explain any five types of operating system.

Q.2 **(A)** Implement using Round Robin Scheduling algorithm. Consider Five Processes P1 to P5 arrived at same time. They have estimated running time 10, 2, 6, 8 and 4 seconds, respectively. Find the average turnaround time and average waiting time for Round Robin (quantum time=3) Scheduling algorithm. Ignore process swapping overhead. **4**

(B) Explain Process states transitions with the help of diagram. **3**

OR

Define Process Control Block (PCB). Explain different fields of Process Control Block (PCB).

(C) Consider three processes (process id 0, 1, 2 respectively) with compute time bursts 2, 4 and 8 time units. All processes arrive at time zero. Consider the longest remaining time first (LRTF) scheduling algorithm. In LRTF ties are broken by giving priority to the process with the lowest process id. Solve the problem using LRTF and calculate average turn-around time. **7**

OR

Consider three processes P1, P2 and P3 with arrival time of 0, 1 and 2 time units respectively, and total execution time of 20, 30 and 40 units respectively. Each process spends the first 30% of execution time doing I/O, then next 60% doing CPU and remaining 10% time doing I/O again. The operating uses a First come first serve scheduling algorithm and schedules a new process either when the running process gets blocked on I/O or when the running process finishes its compute burst. Solve the problem using FCFS and find that for what percentage of time does the CPU remain idle?

Q.3 **(A)** Define Deadlock. Explain necessary conditions for Deadlock. **4**

(B) Explain Producer Consumer problem using Semaphore with algorithm. **3**

OR

Explain Peterson's solution using algorithm.

(C) Explain Readers & Writer Problem with algorithm. **7**

OR

Explain Dinning Philosopher Problem with algorithm.

Q.4 **(A)** Explain paging technique with the help of diagram. **4**

(B) Given six Partition of 300KB, 600KB, 350KB, 200KB, 750KB and 125KB(in order), solve how would the first-fit, best-fit and worst-fit algorithms places processes of size 115 KB, 500KB, 358KB, 200KB and 375KB(in order)? **3**

OR

Consider the snapshot of the system with Five Processes and Four types of resources A,B,C,D.

Allocation				
Process	A	B	C	D
P0	0	0	1	2
P1	1	0	0	0
P2	1	3	5	4
P3	0	6	3	2
P4	0	0	1	4

Max			
A	B	C	D
0	0	1	2
1	7	5	0
2	3	5	6
0	6	5	2
0	6	5	6

Currently available set of resources is (1,5,2,0).
Solve the following Questions using banker's algorithm.
1. Find the content of Need Matrix.
2. Is the System in Safe State?
3. If request from Process P1 arrives for (0,4,2,0) can the request be granted immediately.

(C) Consider the following reference string. Solve and calculate the page fault rates for FIFO page replacement algorithm. Assume the memory size is 4 page frame. **7**

1,2,3,4,5,3,4,1,6,7,8,7,8,9,7,8,9,5,4,5,4,2

OR

Consider the following reference string. Solve and calculate the page fault rates for OPTIMAL page replacement algorithm. Assume the memory size is 4 page frame.

1,2,3,4,5,3,4,1,6,7,8,7,8,9,7,8,9,5,4,5,4,2

Q.5 **(A)** Define RAID. Explain RAID0, RAID1 and RAID2. **4**

(B) Explain different file operations. **3**

OR

Explain different attributes of file.

(C) Consider an imaginary disk with 51 cylinders. A request comes in to read a block on cylinder 11. While the seek to cylinder 11 is in progress, new requests come in for cylinders 1, 36, 16, 34, 9, and 12, in that order. **7**
Starting from the current head position, solve and find the total distance (in cylinders) that the disk arm moves to satisfy all the pending requests, for each of the following disk scheduling Algorithms?
FIFO and SSTF

OR

Consider an imaginary disk with 51 cylinders. A request comes in to read a block on cylinder 11. While the seek to cylinder 11 is in progress, new requests come in for cylinders 1, 36, 16, 34, 9, and 12, in that order.
Starting from the current head position, solve and find the total distance (in cylinders) that the disk arm moves to satisfy all the pending requests, for each of the following disk scheduling Algorithms?
LOOK and Scan.

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